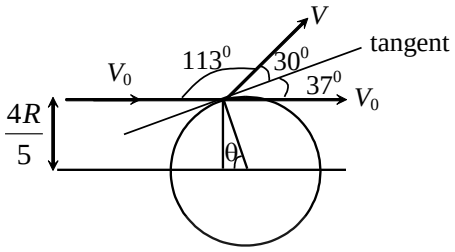
PAPER-I
PHYSICS
SECTION – I
Single Correct Choice Type

1. $\dfrac{1}{R_{eq.}} = \int \dfrac{1}{dR} = \int_{R_1}^{R_2} \dfrac{2Ldx}{\rho\pi x} = \dfrac{2L}{\pi\rho}\ln\dfrac{R_2}{R_1}$

$\therefore \quad R_{eq} = \dfrac{\pi\rho}{2L\ln\left(\dfrac{R_2}{R_1}\right)}$

$\therefore$ **(c)**

2. Since the surfaces are smooth so tangential component of velocity would be conserved.



$$\sin\theta = \frac{4R}{5\times R} = \frac{4}{5}$$

$\theta = 53^0$

So $V_0\cos 37^0 = V\cos 30^0$

or $V_0 \times \dfrac{4}{5} = \dfrac{\sqrt{3}}{2}V$

$8V_0 = 5\sqrt{3}V$

$\therefore$ **(c)**

3. $\sum F_{\text{ext}} = 0$ on 'board and man' system

$$m\times\frac{5l}{2} + m\times 2l = (m+m)x$$

or $x = \dfrac{9l}{4}$

$\therefore$ **(d)**

4. For pure rolling motion

$\alpha.R - 6 = 3$

or $\alpha R = 6 + 3$

$\alpha = \dfrac{9}{3} = 3\,\text{rad/s}^2$

$\therefore$ **(a)**

5. Net charge flowing through switch A is $60\times 3 - 60\times 2 = 60\mu\,\text{C}$

$\therefore$ **(d)**

6. $2A\sin kx = 3\sqrt{2}$

$2\times 3\sin kx = 3\sqrt{2}$

$\sin kx = \dfrac{1}{\sqrt{2}}$

$\dfrac{2\pi}{\lambda}x = \dfrac{\pi}{4};\ \dfrac{3\pi}{4}$

$x = \dfrac{\lambda}{8};\dfrac{3\lambda}{8}$ .....

Distance between consecutive points = $\dfrac{3\pi}{8}-\dfrac{\lambda}{8}=\dfrac{\lambda}{4}$

$\dfrac{\lambda}{4} = 20\,\text{cm}$

$\Rightarrow$ $\lambda = 80$ cm

So, $(n+1)\dfrac{\lambda}{2} = 240$

$\Rightarrow$ $(n+1)\dfrac{80}{2} = 240$

or $n+1 = 6$

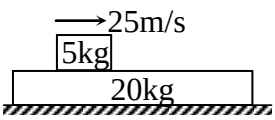
$n = 5$

So string is vibrating in fifth overtone

$\therefore$ **(d)**

SECTION – II
Multiple Correct Choice Type

7. Conserving the momentum



$5\times 25 = 25\times v$

$\Rightarrow$ $v = 5$ m/s

$\Delta K + \Delta U = W_{\text{friction}}$

$\Rightarrow$ $\dfrac{1}{2}(5+20)5^2 - \dfrac{1}{2}\times 5\times 25^2 + 0 = W_f$

$= \dfrac{1}{2}\times 625\times(-4) = -1250\,\text{J}$

$W_{\text{friction}} = -\mu mg\ \ d_{\text{rel}} = -1250$ J

$\Rightarrow$ $\dfrac{5}{8}\times 5\times 10\times d_{rel} = -1250$

$d_{rel} = 40\,\text{m}$

Now, $d_{\text{rel}} = \dfrac{1}{2}a_{rel}.t^2$

$\Rightarrow$ $40 = \dfrac{1}{2}\left(\mu g + \dfrac{\mu mg}{M}\right)t^2$

or $40 = \frac{1}{2} \times 15.625 \times t^2$

$\Rightarrow \quad t = \frac{16}{5} \text{sec}$

$\therefore$ **(a, b, d)**

8. Acceleration of the whole system towards right:

$$a = \frac{F}{M+m}$$

$$F - mg\sin\theta = ma\cos\theta$$

$\Rightarrow \quad F - mg\sin\theta = m\frac{F}{M+m}\cos\theta$

$\Rightarrow \quad F = \frac{(M+m)mg\sin\theta}{M+m-m\cos\theta}$

Pseudo force on m as seen from the frame of M:

$$F_{s1} = ma = \frac{mF}{m+M}$$

$$= mg\sin\theta\left(\frac{m}{M+m(1-\cos\theta)}\right) < mg\sin\theta$$

Pseudo force on M as seen from the frame of m:

$$F_{s2} = Ma = \frac{MF}{m+M}\left(> \frac{mF}{m+M}\right)$$

$$= mg\sin\theta\left(\frac{M}{M+m(1-\cos\theta)}\right) < mg\sin\theta$$

Now $mg\cos\theta - N = ma\sin\theta$

$\Rightarrow \quad N = mg\cos\theta - ma\sin\theta$

Hence N is less than $mg\cos\theta$. Hence it will also be less than $mg\sin\theta$, because $\theta = 45°$.

Applying equation on m in horizontal direction:

$$F\cos\theta - N\sin\theta = ma$$

$\Rightarrow \quad F\cos\theta - N\sin\theta = m\frac{F}{M+m}$

$\Rightarrow \quad N = \frac{mF}{M+m}\left(\frac{(M+m)\cos\theta - m}{m\sin\theta}\right)$

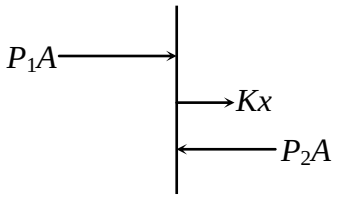
Put $\theta = 45°$

$\Rightarrow \quad N = \frac{mF}{M+m}\left(\frac{M+m-\sqrt{2}m}{m}\right) > \frac{mF}{m+M}$

Normal force between ground and M will be $(M + m)\,g$. It is greater than $mg\sin\theta$. It is also greater than $\frac{mF}{M+m}$, because $\frac{mF}{M+m}$ is less than $mg\sin\theta$.

$\therefore$ **(a, b, c)**

9. The FBD of the piston is



Restoring force = $(P_2 - P_1)A + Kx$

$$-Ma = (P_2 - P_1)A + Kx$$

Also $P_1A(l_0 - x) = PAl_0$

$$P_1 = \frac{Pl_0}{l_0 - x} \quad \ldots \text{(i)}$$

$$P_2A(l_0 + x) = PAl_0$$

$$P_2 = \frac{Pl_0}{l_0 + x} \quad \ldots \text{(ii)}$$

From (i) and (ii) $a = \left[\frac{2Pl_0A}{l_0^2 - x^2} + K\right]\frac{x}{M}$

Since $x << l_0$

$$a = -\left[\frac{2Pl_0A}{l_0^2} + K\right]\frac{x}{M}$$

$\therefore \quad T = 2\pi\sqrt{\frac{Ml_0}{2PA + Kl_0}}$

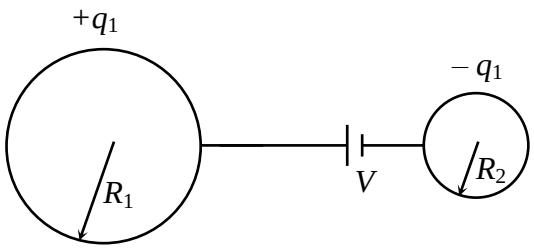
$\therefore$ **(a, b, c)**

10. If only magnetic field is present, the particle will follow a circular path and if only electric field is present, particle will follow a parabolic path.

$\therefore$ **(a, b, c, d)**

11. $\frac{q_1}{4\pi\varepsilon_0 R_1} - \left(-\frac{q_2}{4\pi\varepsilon_0 R_2}\right) = V$

Also number of electrons emitted = $\frac{q_1 - q_2}{e}$



$\therefore$ **(a, b, c, d)**

SECTION–III
Reasoning Type

12. $v_{rel} = \vec{v}_1 - \vec{v}_2$; $\vec{v}_1 = x\sqrt{\frac{K}{M}}$; $\vec{v}_2 = x\sqrt{\frac{K}{2M}}$

$\therefore$ **(c)**

13. $K \propto \frac{1}{l}$ where l is the length of the spring

$\therefore \quad K' = 4K$

$\therefore$ **(a)**

14. $\tan\phi = \dfrac{X_L - X_C}{R}$

If $\phi = 0 \Rightarrow X_L = X_C$

$\therefore \quad Z = R$

$\therefore$ **(b)**

15. The path difference at central point will be zero.

For other fringes, $S_2P - S_1P = n\lambda$ (for constructive interference)

$\therefore$ **(a)**

16. If two particles approach each other under mutual forces, the net force on the system is zero. However, the speed may increase or decrease depending upon the nature of force.

$\therefore$ **(b)**

SECTION–IV
Matrix–Match Type

1. Blocks of mass M_1 and M_2 are connected with a string. Hence their acceleration must be equal.

If the friction force between M_2 and M_3 is sufficient to prevent relative slipping, they move with same acceleration.

$\therefore$ **(A) – (1, 3, 5) ; (B) – (1, 2, 4) ; (C) – (1, 2, 4; D) – (1, 2, 3, 5)**

2. In process *AB*, $P \propto V$.

Also along *AB*, temperature of the gas increases

In process *BC*, volume is constant and pressure increases. Hence temperature will increase.

In process *CD*, pressure is constant and volume decreases. Hence temperature will decrease.

In process *DA*, pressure decreases and volume increases

Also the process *DA* is of the form $P = -mV + C$ where m and C are constants.

Using ideal gas equation

$$\frac{nRT}{V} = -mV + C$$

$$T = -\frac{m}{nR}V^2 + \frac{C}{nR}V$$

Hence temperature first increases then decreases

$\therefore$ **(A) – (1, 3, 4) ; (B) – (1, 2, 4); (C) –2, 5; (D) – 2, 5**

SECTION–V
Integer Answer Type

1. Refraction at plane surface

$u = -(2R + x)$

$$\frac{1}{v} - \frac{\mu}{-(2R+x)} = 0$$

$$v = -\frac{(2R+x)}{\mu}$$

For refraction at curved surface

$$u = -\left(R + \frac{2R+x}{\mu}\right)$$

$$\frac{\mu}{v'} - \frac{1}{-\left(R + \frac{2R+x}{\mu}\right)} = \frac{\mu - 1}{-R}$$

$$v' = \frac{-\mu}{\frac{\mu-1}{R} + \frac{\mu}{2R + R\mu + x}} \qquad \ldots \text{(i)}$$

Now in order to form the final image on O itself this ray should seen to be coming from centre of curvature C.

So $v' = -(R + x)$ … (ii)

From (i) and (ii) and substituting $\mu = 3/2$

$$2x^2 + 9Rx - 8R^2 = 0$$

$$\Rightarrow \quad x = \frac{-9R + \sqrt{145R^2}}{4} = \frac{3R}{4}$$

$x = 6$ cm

Ans.: 6

2. Initially $\omega = \frac{1}{\sqrt{LC_1}} = \frac{1}{\sqrt{400 \times 10^{-6} \times 100}} = 5$ rad/s

$$q = q_0 \cos(\omega t)$$

$$\Rightarrow \quad q = q_0 \cos\left(\frac{\pi}{10} \times 5\right) = 0$$

So at $t = \frac{\pi}{10}$ sec; all the energy of capacitor C_1 would be stored in inductor.

Now, $\omega = \sqrt{\frac{1}{LC_2}} = \sqrt{\frac{1}{100 \times 100 \times 10^{-6}}}$ = 10 rad/s

36% of $\frac{q_0^2}{2C_1} = \frac{q^2}{2C_2}$

$q = \frac{3}{10} q_0$

$\Rightarrow \quad \frac{3q_0}{10} = q_0 \sin(10t)$

$\Rightarrow \quad \sin(10t) = 0.3 = \sin\frac{\pi}{10}$

$t = \frac{\pi}{100}$

Total time $= \frac{\pi}{10} + \frac{\pi}{100} = \frac{11\pi}{100} = \frac{11n\pi}{200}$

So $n = 2$

Ans.: 2

3. Flux of electric field through the intersection area is given by

$\oint \vec{E}.\overrightarrow{ds} = \frac{Qr}{4 \in_0 R^3}(R^2 - r^2)$ where r is distance of sheet from the origin

$= \frac{Q}{8 \in_0 R^2}\left(R^2 - \frac{R^2}{4}\right) = \frac{3Q}{32 \in_0} = \frac{9}{\in_0}$

$x = 9$

Ans.: 9

4. As the wedge accelerates with $\vec{a} = a\hat{i}$ the centre A also has same acceleration.

$\therefore \quad \vec{a}_A = a\hat{i}$

$\vec{v}_A = -v\hat{i}$

$\vec{a}_{B,g} = \vec{a}_{B,A} + \vec{a}_A$ where $\vec{a}_{B,A} = (\vec{a}_{B,A})_t + (\vec{a}_{B,A})_r$ and $\vec{a}_A = a\hat{i}$

$\therefore \quad \vec{a}_{B,g} = \vec{a}_B = (\vec{a}_{B,A})_t + (\vec{a}_{B,A})_r + a\hat{i}$

$(\vec{a}_{BA})_t = \vec{a}_{B_x} - \vec{a}_{A_x} = 0 - a\hat{i}$

$\vec{v}_{B,A} = \vec{v}_B - \vec{v}_A = 2v\hat{i} - (-v\hat{i}) = 3v\hat{i}$

$\therefore \quad (\vec{a}_{B,A})_r = \frac{9v^2}{R}\hat{j}$

$\therefore \quad \vec{a}_{B,g} = -a\hat{i} + \frac{9v^2}{R}\hat{j} + a\hat{i} = \frac{9v^2}{R}\hat{j}$

Putting $v = 3$ and $R = 9$

We get $|\vec{a}_{B,g}| = \frac{9 \times 9}{9} = 9$

$\therefore \quad x = 9$

Ans.: 9